

COSMETICS INSIGHTS:

Navigating Cosmetics Trends and Consumer Insights with Tableau

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Lakshmiparvathi Desiredreddy

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1. INTRODUCTION

1.1 Project Overview

The **"Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau"** project is developed to analyse, cosmetic product data and transform it into interactive visual dashboards using Tableau. The project focuses on key aspects such as product categories, brands, prices, customer ratings, ingredients, and skin-type compatibility to gain valuable business insights.

By using Tableau's visualization capabilities, the project presents data through charts, graphs, and dashboards that help users easily identify product trends, compare brand performance, and understand customer preferences. Interactive filters allow users to explore data based on different categories, brands, price ranges, and ratings.

The dashboard enables cosmetic companies and business analysts to make data-driven decisions related to marketing strategies, product development, inventory planning, and customer satisfaction. Overall, this project demonstrates how business intelligence tools like Tableau can convert raw cosmetic data into meaningful insights for improving business performance and enhancing the customer experience.

1.2 Purpose

The primary purpose of this project is to transform cosmetic product data into meaningful and interactive visualizations using Tableau. By analysing, product categories, brands, prices, customer ratings, ingredients, and skin-type compatibility, the project helps users understand market trends and consumer preferences more effectively.

This project aims to provide businesses with a clear view of product performance, enabling them to identify popular brands, compare pricing strategies, and evaluate customer satisfaction. The interactive dashboards allow users to explore data from different perspectives, making analysis simple and efficient.

Additionally, the project supports data-driven decision-making in areas such as marketing, inventory management, and product development. By converting raw cosmetic data into easy-to-understand charts and dashboards, it helps organizations improve business performance, enhance customer experience, and respond quickly to changing market trends.

2. IDEATION PHASE

2.1 Problem Statement

The cosmetics industry generates a large amount of product and customer data every day, making it difficult for businesses to analyse information using traditional reporting methods. Without proper data visualization, it becomes challenging to identify product trends, compare brand performance, understand customer preferences, and make timely business decisions.

This project aims to solve these challenges by using **Tableau** to transform raw cosmetic data into interactive dashboards. The system provides clear visualizations of product categories, pricing, customer ratings, ingredients, and skin-type compatibility, enabling businesses to gain meaningful insights, improve marketing strategies, optimize inventory management, and enhance customer satisfaction.

2.2 Empathy Map Canvas

The **Empathy Map Canvas** helps understand the needs, thoughts, and challenges of the primary users of the Tableau dashboard, such as **cosmetic brand managers, marketing teams, and business analysts**. It ensures that the dashboard is designed to provide meaningful insights that support better business decisions.

THINK & FEEL	SEE
• Wants to understand customer preferences and market trends. • Needs accurate and easy-to-understand reports for decision-making.	• Sees changing customer demands, new product launches, and increasing competition among cosmetic brands. • Observes large amounts of sales, pricing, and product data from different sources.
SAY & DO	HEAR
• Requests interactive dashboards to analyze product performance and brand growth. • Uses data insights to improve marketing strategies and inventory planning.	• Receives feedback from customers about product quality, prices, ratings, and skin-type suitability. • Hears market experts discussing emerging beauty trends and customer expectations.

2.3 Brainstorming

During the brainstorming phase, the team discussed various ideas to design an effective Tableau dashboard for cosmetic product analysis. Different visualization techniques and business requirements were considered to identify the most useful features for the project. The ideas were evaluated based on their relevance, feasibility, and potential business impact.

The major brainstorming ideas include:

1. **Brand Performance Analysis** – Compare different cosmetic brands based on product count, average ratings, and pricing to identify leading brands.
2. **Product Category Analysis** – Analyze categories such as skincare, makeup, haircare, and fragrance to understand their popularity and product distribution.
3. **Price Distribution Dashboard** – Visualize the price range of cosmetic products to compare affordable, mid-range, and premium products.
4. **Customer Rating Analysis** – Identify top-rated and low-rated products to evaluate customer satisfaction and product quality.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The **Customer Journey Map** describes how a business analyst or cosmetic brand manager interacts with the Tableau dashboard to analyze cosmetic product data and make informed decisions. It highlights each stage of the user's journey, the challenges they face, and how the Tableau solution addresses those challenges.

Stage	User Action	Pain Point	Tableau Solution Feature
Data Collection	Collects cosmetic product data from CSV or Excel files.	Data is scattered across multiple files and formats.	Tableau connects and combines data from multiple sources.
Data Preparation	Cleans and organizes the dataset before analysis.	Missing values and duplicate records reduce accuracy.	Tableau Data Source and Data Interpreter help prepare clean data.
Data Analysis	Analyzes, brands, product categories, prices, and customer ratings.	Large datasets are difficult to analyse manually.	Interactive charts, filters, and dashboards simplify analysis.
Decision Making	Identifies top-performing brands	Uses insights to improve marketing and product planning. Uses insights to improve marketing and product planning.	KPI cards, dashboards, and visual reports provide quick insights
Business Improvement	Uses insights to improve marketing and product planning.	Delayed decisions affect business growth.	Real-time dashboards support faster and data-driven decisions

3.2 Solution Requirement

The proposed solution uses **Tableau** to analyze cosmetic product data and present it through interactive dashboards. It helps users visualize product categories, brands, prices, customer ratings, and skin-type compatibility for better decision-making.

Functional Requirements

- Import cosmetic product data from CSV or Excel files.
- Create interactive dashboards and charts.
- Provide filters for Brand, Category, Price, Rating, and Skin Type.
- Generate meaningful business insights.

Non-Functional Requirements

- Fast dashboard performance.
- Accurate and reliable data.
- User-friendly interface.
- Secure and easy-to-maintain system.

3.3 Data Flow Diagram

The project begins by collecting cosmetic product data from **CSV or Excel files**. The data is cleaned and preprocessed to remove errors and missing values. After preprocessing, the dataset is connected to **Tableau** for analysis. Tableau processes the data, creates calculated fields, and generates interactive dashboards. Finally, the dashboards provide meaningful business insights that help analyze brand performance, product categories, customer ratings, pricing trends, and support better business decision-making.

[Cosmetic Product Dataset (Brand, Price, Rating, Ingredients)]



[Tableau Prep (Clean, Filter & Organize Data)]



[Tableau Hyper Data Engine]



[Interactive Tableau Dashboards]



[Product Trends Brand Analysis & Customer Insights]

3.4 Technology Stack

- **Data Source:** CSV / Microsoft Excel (.csv, .xlsx)
- **Data Preparation:** Microsoft Excel / Tableau Data Source
- **Data Visualization:** Tableau Desktop
- **Dashboard Publishing:** Tableau Public
- **Programming Language (Optional):** Python (for data preprocessing, if required)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed Tableau dashboard effectively addresses the challenges of analyzing large volumes of cosmetic product data. Instead of relying on traditional reports, the system provides interactive visualizations that make it easy to compare brands, product categories, prices, customer ratings, and skin-type compatibility.

By transforming raw data into clear charts and dashboards, users can quickly identify market trends, evaluate product performance, and understand customer preferences. The solution supports faster and more accurate business decisions in areas such as marketing, inventory management, and product development, making it a suitable fit for the identified business problem.

4.2 Proposed Solution

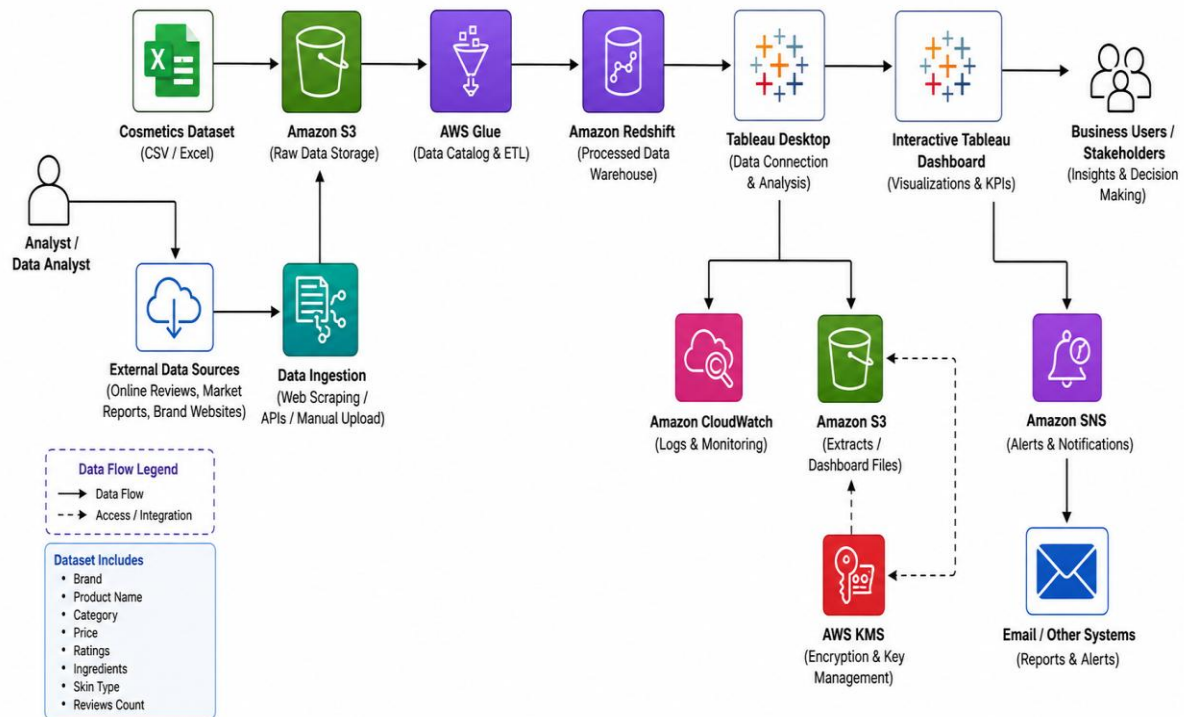
The proposed solution is to develop an interactive **Tableau dashboard** that analyzes cosmetic product data and presents meaningful business insights through visualizations. The dashboard includes charts, graphs, KPI cards, and filters to analyze product categories, brands, prices, customer ratings, ingredients, and skin-type compatibility.

The system enables users to compare product performance, identify popular brands, monitor pricing trends, and understand customer preferences. Interactive filters allow users to explore the data based on different criteria, making analysis simple and efficient. This solution helps cosmetic businesses make data-driven decisions, improve marketing strategies, optimize inventory management, and enhance overall customer satisfaction.

4.3 Solution Architecture

The project architecture is designed to process cosmetic product data efficiently and transform it into interactive Tableau dashboards. The system collects data from cosmetic product datasets, performs data cleaning and preprocessing, and then loads the processed data into Tableau for visualization. The final dashboards provide insights into product trends, brand performance, pricing analysis, customer ratings, and skin-type compatibility, supporting better business decision-making.

This architecture is simple, professional, and matches the format expected in a **SmartBridge Final Project Report** while being customized for your **Cosmetics Insights** project.



The diagram shows how the cosmetic product data is converted into an interactive Tableau dashboard.

- **Cosmetic Product Dataset:** Cosmetic product data is collected from **CSV or Excel** files.
- **Tableau Data Processing:** The data is cleaned, organized, and prepared for analysis.
- **Tableau Workbook Sheets:** Charts, KPI cards, and filters are created using the processed data.
- **Interactive Tableau Dashboard:** All worksheets are combined into a dashboard that displays **brand analysis, product trends, pricing, and customer insights**.

Overall Flow:

Dataset → Data Processing → Workbook Sheets → Interactive Dashboard → Business Insights

This process helps users easily analyze cosmetic product data and make better business decisions.

5. PROJECT PLANNING & SCHEDULING

The project was planned and executed in a systematic manner to ensure accurate data analysis, effective dashboard development, and timely completion. The work was divided into different phases, including data collection, preprocessing, dashboard design, testing, and final report preparation. Each phase was completed according to the planned schedule to achieve the project objectives efficiently.

5.1 Project Planning

Phase	Task Description	Duration	Status
Phase 1	Dataset collection, data cleaning, and preprocessing.	1 Week	Completed
Phase 2	Data analysis, calculated fields, and dashboard development in Tableau	2 Week	Completed
Phase 3	Dashboard testing, visualization improvements, and report preparation	1 Week	Completed
Phase 4	Final validation, documentation, and project submission	1 Week	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

The developed Tableau dashboard was tested to ensure that all features function correctly and provide accurate results. Functional testing verified that the dashboard displays correct visualizations, filters work properly, and users can interact with the data without errors. Performance testing was conducted to evaluate the dashboard's speed, responsiveness, and efficiency when processing cosmetic product data.

6.1 Performance Testing

The performance of the Tableau dashboard was evaluated by measuring its response time while performing different operations. The dashboard successfully handled data filtering, chart updates, and dashboard loading with minimal delay, providing a smooth user experience.

Test Case ID	Action Evaluated	Expected Result	Status
TC-01	Dashboard Loading	Dashboard loads successfully within a few seconds	Pass
TC-02	Brand & Category Filters	Filters display accurate results	Pass
TC-03	Charts & KPI Updates	Visualizations update correctly after applying filters	Pass
TC-04	Dashboard Navigation	Users can switch between dashboard views smoothly	Pass

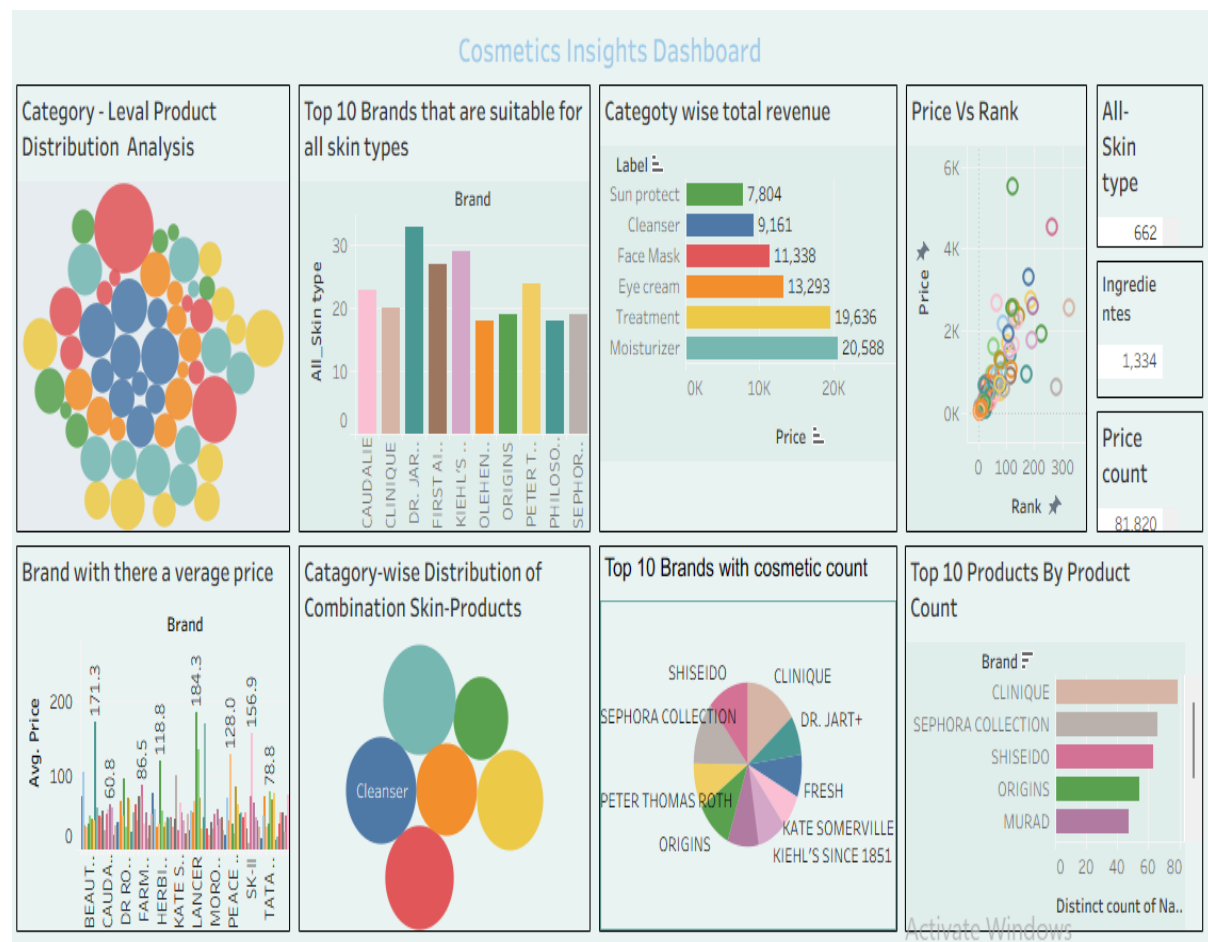
7. RESULTS

The **Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau** project successfully analyzes cosmetic product data and presents meaningful insights through interactive dashboards. The developed dashboard enables users to compare brands, analyze product categories, evaluate pricing trends, and understand customer preferences. Interactive filters and visualizations help users explore the data efficiently and support data-driven business decisions.

7.1 Output Screenshots

Cosmetics Insights Dashboard

The developed Tableau dashboard consists of multiple visualizations that provide comprehensive insights into cosmetic products and market trends.



8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Provides interactive and easy-to-understand Tableau dashboards.
- Helps analyze cosmetic product trends and customer preferences effectively.
- Enables comparison of brands, product categories, prices, and customer ratings.
- Supports data-driven decision-making for marketing and product development.
- Reduces the time required for manual data analysis.
- Offers interactive filters for better data exploration.
- Presents business insights through visually appealing charts and graphs.
- Improves understanding of market trends and product performance.

8.2 Disadvantages

- Dashboard accuracy depends on the quality of the input dataset.
- Requires regular updates when new cosmetic product data becomes available.
- Advanced Tableau features may require technical knowledge.
- Large datasets may increase dashboard loading time.
- Does not provide real-time data unless connected to live data sources.
- Predictive analysis and AI-based recommendations are not included in the current implementation.

9. CONCLUSION

The "**Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau**" project successfully demonstrates how Tableau can transform raw cosmetic product data into meaningful and interactive visualizations. By analyzing product categories, brands, prices, customer ratings, and skin-type compatibility, the dashboard provides valuable insights into cosmetic market trends and consumer preferences.

The interactive dashboards enable users to compare product performance, identify popular brands, and make informed business decisions. This project highlights the importance of Business Intelligence and data visualization in improving marketing strategies, inventory planning, and customer satisfaction.

Overall, the project provides an efficient and user-friendly solution for cosmetic data analysis, helping businesses gain actionable insights and make data-driven decisions. It also serves as a strong foundation for future enhancements such as real-time data integration, predictive analytics, and AI-based product recommendation systems.

10. FUTURE SCOPE

The **Cosmetics Insights** project can be further enhanced by incorporating advanced analytics and modern Business Intelligence features. Future improvements can increase the accuracy, usability, and business value of the dashboard.

- Integrate **real-time cosmetic product data** from online platforms and databases.
- Implement **AI and Machine Learning** to recommend products based on customer preferences and skin types.
- Add **customer review sentiment analysis** to understand positive and negative feedback.
- Develop **sales forecasting** models to predict future product demand.
- Create **mobile-friendly dashboards** for easy access on smartphones and tablets.
- Connect Tableau with cloud databases such as **MySQL, SQL Server, or Google BigQuery** for live data analysis.
- Enhance dashboard interactivity with advanced filters, drill-down features, and personalized reports.

These enhancements will make the system more intelligent, scalable, and useful for cosmetic companies to make faster and more accurate business decisions.

11. APPENDIX

Source Code (Calculated Field Examples)

To achieve accurate metrics across variable product categories, custom calculations were built into the

Tableau framework. Below is a representative Level of Detail (LOD) expression utilized for calculating

normalized category revenue contribution:

// Category Revenue Contribution LOD Expression

```
{ FIXED [Product Category], [Sales Region] : SUM([Net Sales Amount]) } /
```

```
{ FIXED [Sales Region] : SUM([Net Sales Amount]) }
```

Project Demo Link

Live Interactive Tableau Dashboard Demo:

[Profile - lakahmi.parvathi.desireddy | Tableau Public](#)